

JavaScript Fundamentals

Student Assignment Sheet · Arrays · Loops · Functions

● 3 Assignments

● Beginner → Intermediate

● Watch · Code · Submit



Before each assignment, watch the required video sections

Timestamps below link directly to the right part. Watching is required to complete each task.

■ 00:34:44 Arrays

■ 00:42:06 Destructuring

■ 00:44:25 While Loops

■ 00:45:21 For Loops

■ 00:59:58 Functions

■ 01:03:13 Map · Filter · Reduce

Assignment 1 — Your First Array: Grade Tracker

01 / ARRAYS

Topic: Arrays · Array Methods · Destructuring · Spread Operator

Watch first →

■ 00:34:44 Arrays

■ 00:42:06 Destructuring

SCENARIO

You are building a simple grade tracker for a class of 5 students. Each student has a name and a score out of 100. Store this data using JavaScript arrays and manipulate it using the array methods you watched in the video.

TASKS — complete all of these

- ✓ Create an array called 'students' that holds 5 student names as strings
- ✓ Create a second array called 'scores' that holds their matching scores as numbers
- ✓ Use .push() to add a new student and score to both arrays
- ✓ Use .pop() to remove the last student from the students array and log what was removed
- ✓ Use array destructuring to pull the first two student names into their own variables and log them
- ✓ Use the spread operator (...) to combine both arrays into one new array called classData

STARTER CODE

```
// Assignment 1 — Grade Tracker

const students = ["Amara", "Tunde", "Bisi", "Chidi", "Ngozi"];
const scores    = [78, 85, 90, 62, 95];

// 📄 Your code goes below
```

■ Bonus Challenge

Use .indexOf() to find a student's position by name, and .includes() to check whether a score of 100 exists in the scores array.

Assignment 2 — Loop It: FizzBuzz & Class Report

02 / LOOPS

Topic: For Loops · While Loops · Conditionals

Watch first →

■ 00:44:25 While Loops

■ 00:45:21 For Loops

SCENARIO

You need to loop through the student grade data from Assignment 1 and generate a report card. You will also tackle the classic FizzBuzz interview question — every developer should know this one.

PART A — CLASS REPORT CARD

Using a for loop, iterate through students and scores and print: "Amara scored 78" for each student

Inside the same loop, add an if/else to print a grade letter: A (90+), B (75-89), C (60-74), F (below 60)

Maintain a running total variable and calculate the class average after the loop ends

PART B — CLASSIC FIZZBUZZ

Write a for loop from 1 to 30: print "Fizz" if divisible by 3, "Buzz" if divisible by 5, "FizzBuzz" if both, otherwise the number

Rewrite the exact same FizzBuzz using a while loop instead of a for loop

STARTER CODE

```
// Assignment 2 — Loops (reuse your arrays from Assignment 1)

const students = ["Amara", "Tunde", "Bisi", "Chidi", "Ngozi"];
const scores   = [78, 85, 90, 62, 95];
let total      = 0;

// Part A — report card for loop

// Part B — FizzBuzz with for loop

// Part B — FizzBuzz with while loop
```

■ Bonus Challenge

Use a for...of loop to print only the names of students who scored above the class average. Hint: calculate the average first, then loop a second time.

Assignment 3 — Functions & Higher-Order Magic

03 / FUNCTIONS

Topic: Functions · Arrow Functions · .map() · .filter() · .reduce()

Watch first →

■ 00:59:58 Functions

■ 01:03:13 Map · Filter · Reduce

SCENARIO

You're now a developer at a school and you need to write reusable functions that process student data. You'll use JavaScript's most powerful array tools: .map(), .filter(), and .reduce() — the tools professional developers use every single day.

PART A — WRITE REUSABLE FUNCTIONS

Write a function called getGrade(score) that takes a score and returns the letter grade (A, B, C, or F)

Write an arrow function version of the same getGrade and assign it to a const variable

Write a function called announce(name, score) that returns a string like: "Amara — Grade: B (85)"

PART B — MAP, FILTER & REDUCE

Use .map() and your getGrade function to transform the scores array into a new grades array of letter grades

✓ Use .filter() to create a new array containing only the students who passed (score ≥ 60)

Use .reduce() to calculate the total of all scores — no for loop allowed!

Chain .filter() and .map() in one statement to get only the names of students who earned an A grade

STARTER CODE

```
// Assignment 3 — Functions & Higher-Order Methods

const students = ["Amara", "Tunde", "Bisi", "Chidi", "Ngozi"];
const scores = [78, 85, 90, 62, 95];

// Part A — regular function
function getGrade(score) {
  // your code here
}

// Part A — arrow function version
const getGradeArrow = (score) => {
  // your code here
};

// Part B
const grades = // TODO: .map()
const passed = // TODO: .filter()
const totalScore = // TODO: .reduce()
const aStudents = // TODO: .filter() + .map() chained
```

■ Bonus Challenge

Combine all three methods in one chain — `.filter().map().reduce()` — to compute the average score of passing students only, with no extra variables.

■ How to Submit

- Save each assignment as a separate .js file: `assignment1.js`, `assignment2.js`, `assignment3.js`
- Run your code in Node.js (■ 00:05:40 in the video) and confirm there are no errors before submitting
- Include comments in your code explaining what each section does — this shows you understood it
- Bonus tasks are optional but earn extra marks
- Assignments build on each other — complete in order: 1 → 2 → 3